

SKILL SESSION • 90 MIN •
HANDS-ON

Mastering the Scientific Narrative

*A Practical Guide to Reading and Writing High-Impact AI
Research Papers*

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Theme: Sovereign Intelligence



Dr. Ing. Castro G. Hounmenou



RESOURCE KIT

Scan now — templates, checklist &
reading list are live from this first
slide.



YOUR FACILITATOR

Who is standing in front of you?



Dr. Ing. Castro G. Hounmenou

Assistant Professor of AI, Université de Labé ·
Post-doc, CERFIG



PhD, Statistics-Probability

Machine Learning & Data Mining. 15+ years leading data-collection and statistical analysis projects across West Africa.



50+ indexed publications

Machine learning, epidemiology and agro-climatology — including work shown at NeurIPS workshops.



World Top Scientist 2024

In Biostatistics. Best Researcher Award 2025 in Infectious Disease Modeling.



Guinea's AI Strategy

Member of the national Scientific Committee — and a reviewer who has read hundreds of submissions.

Why this matters: I sit on both sides of the review process — today you get what I look for when I read your paper.





Research that is not written does not exist. Research that is not published does not travel.

AI research is not only about writing code. It is about establishing African scientific agency — documenting and publishing local innovations where the global community reads them.

NeurIPS

ICLR

CVPR

Nature MI

AfricaNLP

Indaba

Sovereign Intelligence → **Sovereign Narrative.** Owning our stories is how African-led solutions get peer-reviewed, cited, and built upon.

Resource Kit →





OPENING • ROADMAP

Our 90 minutes together

1

Opening & context

5'

2

Anatomy of an AI
paper

8'

3

3-Pass reading
method

14'

4

The Writing Blueprint

18'

5

Resource Kit + Q&A

7'

6

Write your abstract in
Overleaf

23'



Superpower 1 — Read efficiently

Deconstruct any dense AI paper with the systematic 3-Pass method.



Superpower 2 — Write with impact

Build a scientific story that meets NeurIPS / ICLR reviewer standards.

+ 15 extra minutes of open discussion at the end — and the Resource Kit is already live on the QR code, from slide 1.

Resource Kit →





The structural DNA of a modern AI paper

Classic science (IMRAD)

Introduction

Methods

Results

And

Discussion



Modern ML style (NeurIPS / ICLR / CVPR)

Title + Abstract *the paper in 200 words — 80% of readers stop here*

Introduction *problem, gap, contributions (often as a bullet list)*

Related Work *position yourself in the conversation*

Methodology *your model / approach, with clear notation*

Experiments & Results *datasets, baselines, metrics, ablations*

Discussion • Limitations • Broader Impact *honesty builds trust*

Same skeleton, new muscles: ML papers add Related Work, ablations and Broader Impact.





PART 1 • THE ANATOMY OF AN AI PAPER

The hidden logic — every section answers one question

Abstract

“Why should I spend 10 minutes on this paper?”

Introduction

“What problem, what gap, what is new here?”

Related Work

“How is this different from what already exists?”

Methodology

“Could I re-implement this myself?”

Results

“What is the evidence — and is it fair?”

Limitations

“Where does this honestly stop working?”

Reading tip: before reading a section, say its question out loud. Writing tip: if your section does not answer its question, re-

Resource Kit →





Visual map — page 1 of a NeurIPS paper



1

Title — a promise in ≤ 15 words

2

Authors & affiliations — your African lab on the map

3

Abstract — the paper in miniature (our exercise later!)

4

Figure 1 — often summarizes the whole idea; readers jump here

5

Contributions list — reviewers hunt for this bullet list

Live demo: we open a real annotated paper (African AI example) and locate 1→5 together.

Resource Kit →





The beating heart: the contributions list

At the end of every strong Introduction, reviewers expect an explicit list:

“Our contributions are threefold:”

- A new method — we propose X, the first ... for low-resource settings;
- A new resource — we release a dataset of N samples collected in ...;
- New evidence — X outperforms strong baselines by Y% on Z benchmarks.



Reviewer's lens

Each claimed contribution must be verifiable in the Results section. Claim 3 things, prove 3 things.

Types of contribution: method • dataset/resource • benchmark • analysis/theory • open-source tool.





A real abstract, dissected

AgroNet — Abstract (the paper you just saw)

Cassava feeds over 500 million Africans, yet mosaic and brown-streak diseases destroy up to 40% of yields. Existing vision models for plant diagnosis require cloud connectivity that is absent in most rural areas. We introduce AgroNet, an offline, quantized convolutional network optimized for low-end mobile devices. Trained on 12,406 leaf images collected across 42 farms in Guinea and Nigeria, AgroNet reaches 78.3% top-1 accuracy while cutting inference latency by 40% versus MobileNetV3. We release our dataset, code and models to support agricultural AI across the continent.

The 5 moves of a great abstract



CONTEXT — stakes & numbers



GAP — what blocks the world



METHOD — your one-line answer



RESULTS — data + hard numbers



IMPACT — the door you open

Our 3-sentence exercise compresses these 5 moves into 3: Context+Gap → Method → Results+Impact.





Inside the mind of a reviewer

What gets scored



Clarity of the core contribution

can I state it in one sentence?



Novelty

what is genuinely new vs. related work?



Rigor & reproducibility

protocol, baselines, code, seeds



Quality of evidence

fair comparisons, honest error bars

Common desk-rejection causes



Missing or weak baselines



Unclear notation, undefined symbols



Overclaiming (“solves”, “first ever”)



Unreadable figures, no axis labels



Mini-quiz (show of hands): I read two abstracts aloud — A (vague, overclaiming) vs B (precise, evidence-based). Which one would YOU accept?





From a field in Benin to a NeurIPS poster

A real African paper, told properly — this is narrative sovereignty in practice.

1

The local question

Climate change is cutting maize yields in Benin, and farmers cannot control the weather. Which climatic parameters actually drive a high yield?

2

The African dataset

26 years (1995–2020) of climate and maize-yield data, 5 districts with synoptic stations, across two agro-climatic zones — Sudanian and Sudano-Guinean.

3

The honest result

A decision tree fits almost perfectly ($R^2 = 0.998$) but nobody can read it — so the team mined frequent patterns to produce rules a farmer can act on.

4

The global venue

Presented at a NeurIPS 2022 workshop, then published in IJ-AI (2024). Local problem → global conversation.

Read the paper: Tahi, Houndji, Hounmenou & Glèlè Kakaï — IJ-AI 13(4):3930–3941, 2024. Full reference in your kit.





A published African abstract, move by move

Odounfa, Hounmenou et al. — *Discover Artificial Intelligence*, 2025

Chili pepper is a major crop in West Africa, but anthracnose and TYLCV severely cut its yields and farmers' income. Visual inspection is slow, labour-intensive and depends on individual expertise. We built a dataset of 534 leaf images collected in Benin under varied lighting, angles and backgrounds, and compared 13 deep-learning models. YOLOv8 reaches $\text{mAP}@0.5 = 0.995$ while MobileNetV2 and DenseNet121 reach 96.25% accuracy; YOLOv8 runs in 1.2 ms per image with only 3.1M parameters. Deep learning can therefore support real-time, automated disease detection for smallholder farmers.

1

CONTEXT

the crop, the region, the stakes

2

GAP

why today's practice fails

3

METHOD

the dataset + what was compared

4

RESULTS

named metrics, exact numbers

5

IMPACT

the door it opens

Notice: no adjective does any work. Every claim is a number a reviewer can check, against a named baseline.



Why we read badly

The linear, page-by-page approach fails on math-heavy ML papers with 30-page appendices. You run out of time and confidence before reaching the point.



Page 1 → page N

reading like a novel



Every equation

*decoding all proofs on first
contact*



No decision point

*never asking “is this worth my
time?”*

The fix: S. Keshav's “3-Pass” method (2007) — read a paper in up to three deliberate passes, each with a goal, a duration, and an exit door.





Pass 1 — The quick scan

Read **ONLY**: title • abstract • introduction • section headings • figures • conclusions.

Then answer the 5 C's:



Category

What type of paper is it?
(method, benchmark, survey...)



Context

Which prior work does it build
on?



Correctness

Do the assumptions look valid?



Contributions

What are the claimed
novelties?



Clarity

Is it well written?



Exit door: after Pass 1 you **DECIDE** — discard, keep for reference, or go deeper. Most papers legitimately stop here. This is also how reviewers and busy PIs will read YOUR paper.





Pass 2 — Grasping the content

- 1 **Read with care, skip the proofs** — *note key theorems, ignore heavy derivations for now*
- 2 **Interrogate every figure & table** — *axes, units, error bars — is the evidence convincing?*
- 3 **Trace the experimental setup** — *datasets, baselines, metrics, hyperparameters*
- 4 **Take structured notes in the margin** — *your future self is your first reader*
- 5 **Mark the references you must read next** — *this grows your literature map*



Success test

You pass Pass 2 when you can summarize the paper's main idea, evidence and limits to a colleague — without opening it.

It is normal not to understand everything — jot questions; some resolve in Pass 3, some become YOUR research ideas.





Pass 3 — Critical reconstruction

Virtually re-implement the paper: re-derive the approach with the same assumptions, then compare your reconstruction with the authors' choices.



Rebuild

Re-derive the method step by step; challenge every design choice — “why a transformer here and not a CNN?”



Expose

Identify hidden assumptions, missing baselines, weak statistics, and failure cases the authors did not test.



Project

Ask: “What would I have done differently? What experiment is missing?” — this is where new papers are born.

When to invest in Pass 3: papers central to your own research, literature reviews, and when you serve as a peer reviewer.



LIVE DEMO • 4 MIN

Watch my eyes: Pass 1 on a real paper

Demo paper (African AI context — printed excerpt in your hands):

e.g., a Masakhane NLP paper, or an AI-for-drought-prediction / crop-disease-detection paper

1

Title + abstract

~60 sec

2

Headings + Figure 1

~90 sec

3

Conclusions

~60 sec

4

Verdict: the 5 C's out loud

~60 sec

A visible timer runs on screen. Follow along on your printed copy — you'll do this yourselves in a moment.

Resource Kit →





What just happened in 4 minutes?



We answered the 5 C's

category, context, correctness, contributions, clarity — from ~10% of the text



We made a decision

keep / discard / go deeper — reading time is a budget you allocate



We never read the method

and yet we can already discuss the paper intelligently

Your weekly practice

- 3 papers / week in Pass 1
- 1 paper / week in Pass 2
- 1 paper / month in Pass 3

≈ 90 min per week, total.

Reading is a budgeted skill: minutes in, decisions out.





The 3-Pass method at a glance

Pass	Goal	Time / paper	Output
1 • Scan	Triage — get the bird's-eye view (5 C's)	5–10 min	Decision: discard / keep / deepen
2 • Grasp	Understand content, evidence, figures	up to 1 h	Summary you can tell a colleague
3 • Reconstruct	Virtually re-implement; critique deeply	1–5 h	Strengths, weaknesses, new ideas

Tools that multiply the method

Zotero / Mendeley

reference managers

Notion / Obsidian

structured reading notes

Elicit • Scite.ai

AI-assisted discovery — verify, never outsource judgement

Resource Kit →



Building your story in four steps

A paper is not a diary of what you did — it is a story engineered for a busy, skeptical reader.



1 • Frame

Title & Introduction that demand attention



2 • Core

Methodology, notation & FAIR data



3 • Impact

Results & publication-quality figures



4 • Honesty

Discussion, limitations, conclusion

Each step maps to a section of the Scientific Writing Checklist you take home.

Resource Kit →





Titles: your 15-word promise

Formulas that work

[Method]: [What it does] for [Problem]

"AgroNet: Lightweight Crop-Disease Detection for Offline Mobile Use"

[Question or claim as hook]

"Do Multilingual Models Understand West African Languages?"

[Resource] + [scale/context]

"NaijaVoices: a 500-Hour Speech Corpus for Nigerian Languages"

Traps to avoid

Vague & grand

"A Novel Approach to Artificial Intelligence in Africa"

Keyword-free

reviewers & search engines can't find or classify it

Overclaiming

"Solving Healthcare with Deep Learning" — no.

Test: could a stranger guess your method, problem AND context from the title alone?





The Introduction funnel

CONTEXT — the world

Crop diseases threaten food security across West Africa...

GAP — what's missing

...but existing vision models need cloud compute unavailable to rural farmers.

CONTRIBUTION — you

We introduce AgroNet, an offline, quantized detector for low-end phones.

Heilmeier Catechism (extract)

- What are you trying to do? (no jargon)
- How is it done today, and what are the limits?
- What is new in your approach?
- Who cares — locally and globally?

Answer these 4 → your Introduction writes itself.

General → specific → YOU. End with the contributions list (slide 8).





Before / after: the same research, two stories



BEFORE — activity report

"In this work we used a CNN to classify images of cassava leaves. We collected data and trained the model with Keras. Deep learning is a powerful technique that has many applications. Our results are presented below."

No stakes, no gap, no claim — the reader has no reason to continue.



AFTER — scientific story

"Cassava feeds 500M Africans, yet mosaic disease destroys up to 40% of yields [context]. Existing detectors require cloud connectivity absent in rural areas [gap]. We introduce an offline detector that reaches 91% accuracy on-device, enabling diagnosis where it matters [contribution]."

Stakes → gap → measurable claim. Same project, publishable frame.

Notice: the 'after' version is also the skeleton of a perfect abstract — keep it in mind for the exercise.





Methodology: write for re-implementation

The gold standard: a competent reader could rebuild your system from your text alone.

1

Consistent notation

scalars x , vectors $\mathbf{x}$ (bold), matrices $\mathbf{X}$ (bold caps); define EVERY symbol at first use — one notation table saves reviewers.

2

Architecture & training

layers, losses, optimizer, learning-rate schedule, hardware; hyperparameters in a table, search space stated.

3

Evaluation protocol

metrics with formulas, train/val/test splits, number of seeds, and the baselines you compare against — chosen fairly.



Reviewer heuristic: a Methodology I cannot re-implement is a Methodology I cannot trust — and cannot accept.





Your dataset deserves a full biography

African datasets are scientific treasures — describe them like one:

Provenance

where, when, by whom, with what consent and ethics approval

Size & composition

N samples, class balance, languages/regions covered

Preprocessing

cleaning, filtering, augmentation — every transformation stated

Splits & leakage control

how train/val/test were separated (by farm, by speaker...)

Known biases

urban vs rural coverage, device quality, seasonal effects

Access & license

where others can find it and under which terms

Best practice: publish a “dataset card / datasheet” alongside the paper — template included in your kit.





FAIR data — turning local data into global assets



F • Findable

DOI + rich metadata; indexed on Zenodo, HuggingFace Hub, or an institutional repository.



A • Accessible

clear retrieval protocol — open where possible, controlled where ethics requires (e.g., health data).



I • Interoperable

standard formats (CSV, JSON, HF datasets), documented schemas, standard vocabularies.



R • Reusable

explicit license (CC-BY, CC-BY-NC...), dataset card, provenance — so others can build on it, citing YOU.

Sovereignty in practice: a FAIR African dataset gets cited, benchmarked, and becomes infrastructure the global community depends on — with African authors on every citation.





Reproducibility: your repo is part of the paper

agronet/

— README.md	← start here
— requirements.txt	← exact versions
— data/	← or download
script	
— src/	← clean,
commented	
— configs/	← all hyperparams
— scripts/run.sh	← one-command
repro	
— results/	← logs & figures

Pin everything

library versions, random seeds, hardware noted in the paper

One command to reproduce

if a reviewer can run run.sh, you've won their trust

Link it

anonymized repo for review; public + archived (Zenodo DOI) at camera-ready

Reproducibility is doubly strategic for African labs: it neutralizes skepticism and showcases engineering quality.





Choose the chart that carries your claim

Line curves

evolution & convergence

training loss, accuracy vs. epochs, scaling laws

Bar charts

comparing methods

your model vs. baselines on each benchmark

Confusion matrix

where the model fails

per-class errors — vital in health applications

Scatter / Pareto

trade-offs

accuracy vs. latency on low-end devices

Box plots

variance across seeds

honest reporting: mean $\pm$ std over ≥ 3 runs

Qualitative panels

showing, not telling

example predictions, attention maps, error cases

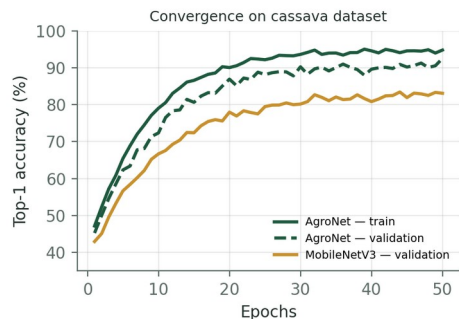
One figure = one message. If you must explain a figure for 2 minutes, it is the wrong figure.



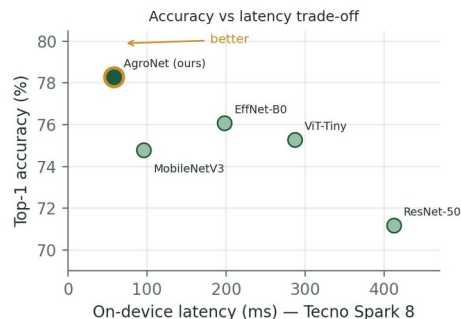


PART 3 • STEP 3 — IMPACT

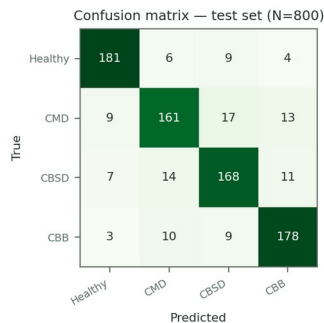
The four figures of a strong Results section



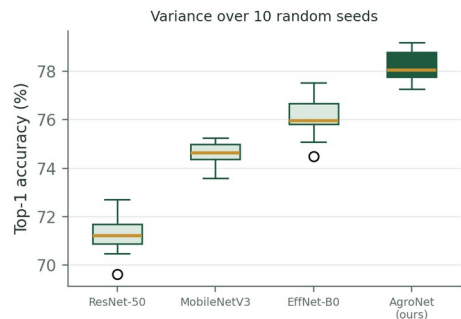
① Convergence — trains stably, beats the baseline



② Trade-off — best accuracy at lowest latency: the low-resource argument in one image



③ Confusion matrix — “here is exactly where we fail” (vital in health & agri)



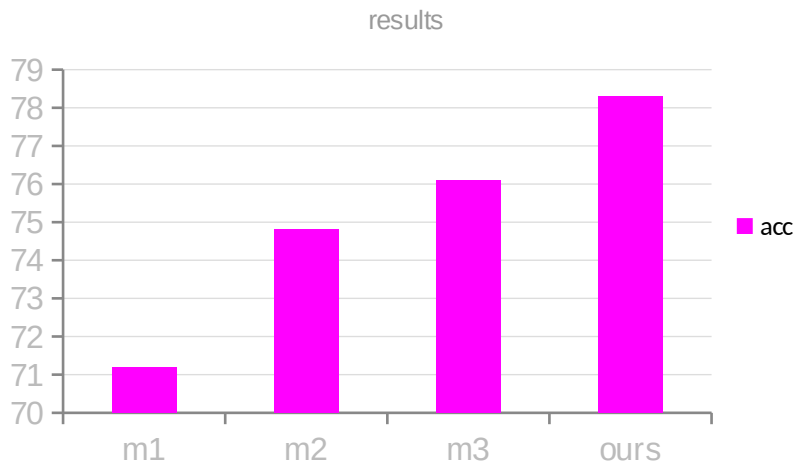
④ Seed variance — the gain is not luck: 10 seeds, honest spread





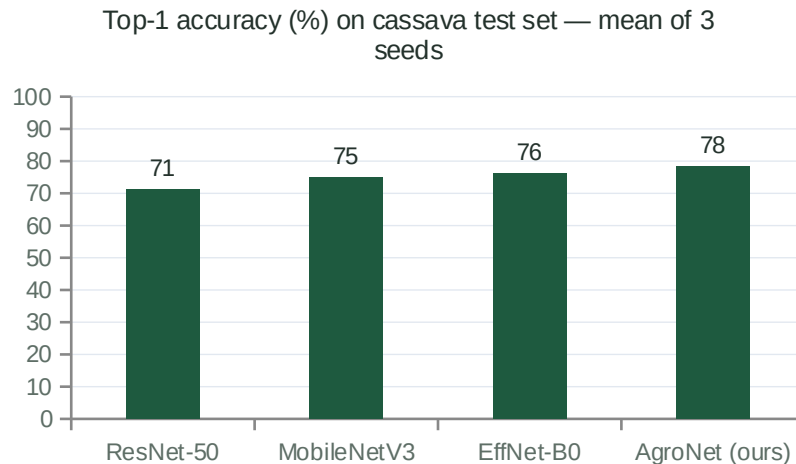
Spot the errors — vote A or B!

Figure A



Truncated axis exaggerates gains • cryptic labels • no units • no error bars

Figure B



Full axis • named baselines • units & protocol in caption • honest comparison

Same numbers — different credibility. Reviewers see hundreds of Figure A's.





Publication-quality checklist for every figure



Self-contained caption

a reader should understand the figure without the body text — state what, where, and the takeaway.



Labeled axes with units

“Accuracy (%)”, “Latency (ms)”, “Epochs” — never bare numbers.



Readable at print size

fonts ≥ 8 pt after scaling; test by zooming to 100% on one column.



Honest scales & colors

no truncated axes without a break mark; colorblind-safe palettes; consistent colors across figures.



Vector graphics

export PDF/SVG from matplotlib or TikZ — never blurry screenshots.



Compare against baselines

your bar never stands alone; context creates credibility.

Tables: same rules — booktabs style, bold your best result, report mean $\pm$ std.



“Better” is not a result — it is a hypothesis

Two accuracies differ. So what? Reviewers ask whether the gap survives a statistical test.

1 Report a distribution, not a point

Run k-fold cross-validation or ≥ 3 seeds and report mean $\pm$ std. A single lucky run is not evidence.

2 Test before you claim

Check normality (Shapiro-Wilk). If it fails — as it usually does with small African field datasets — use Kruskal-Wallis, not ANOVA.

3 Then say which pair differs

A significant omnibus test says “somebody differs”. A post-hoc test (Nemenyi, Dunn) says who. Report the critical difference.

4 Report the trade-off, not just the score

Accuracy, parameters, GFLOPs, inference time. On a \$50 phone, 1.2 ms at 3.1M parameters IS the contribution.

This is exactly what the chili-pepper paper does — and it is the single cheapest way to look like a serious lab.





Discussion, Limitations & Conclusion

Discuss — don't repeat

Interpret: WHY does your method win? When does it not? What does it mean for the field and for deployment?

Limitations — the trust section

Honest, specific, self-critical. E.g.: “AgroNet was validated on 3 cassava varieties; performance under extreme low-light and on 2G-only devices remains untested.” Reviewers respect this — hiding limits is how papers die in review.

Conclusion — land the story

One paragraph: restate contribution + strongest number + future work that follows naturally from your limitations.





PART 4 • RESOURCE KIT HANDOVER

Your Resource Kit — everything on one QR code

1

Scientific Writing Checklist

step-by-step pre-submission guide covering all four blueprint steps

2

LaTeX / Overleaf templates

pre-configured for NeurIPS, ICLR and major venues — start writing today

3

Curated annotated reading list

Keshav's 3-Pass paper, FAIR principles, writing guides, African AI venues

4

Tools & templates

Zotero/Mendeley guides, dataset-card template, reproducible-repo skeleton



Scan → GitHub repo
[github.com/Castro2026/
DeepLearningIndaba2026-
Scientific-Narrative-Academy](https://github.com/Castro2026/DeepLearningIndaba2026-Scientific-Narrative-Academy)

Open the Overleaf template NOW — you will write your abstract straight into it. Printed backups and a short URL on the board for anyone without a smartphone.





Abstract makeover — 60 seconds before you write



BEFORE

"Malaria is a big problem in Africa. In this project we used machine learning and deep learning to detect malaria. We tested our model and the results were very good and promising. This shows AI can help healthcare."

No numbers • no gap • "very good" is not evidence • could describe 10,000 projects



AFTER — 3 sentences

"Rural clinics in West Africa diagnose malaria by manual microscopy, taking 30+ minutes per slide with scarce experts. We introduce PaluNet, a compact detector that runs on a \$50 smartphone attached to any microscope. On 8,500 field slides from 6 clinics, PaluNet reaches 94% sensitivity in under 20 seconds, matching expert microscopists."

Context+gap → method → numbers. 3 sentences, fundable.

Your turn now — same transformation, your scenario.



Write your 3-sentence abstract

S1 • CONTEXT / PROBLEM

"In African agricultural zones, crop disease X causes major yield losses, but existing vision models require high-compute resources."

S2 • METHOD

"We introduce [Model], an offline, quantized network optimized for low-end mobile devices."

S3 • IMPACT / RESULTS

"Our approach reaches X% accuracy on local data while cutting inference time by Y% versus baselines."

Pick ONE scenario card:



Climate — flood forecasting



Agriculture — pest detection



Health — malaria diagnosis



NLP — local languages

Solo or in pairs • write straight into your Overleaf copy • take the time you need • volunteers read theirs aloud for live feedback.

Judge yourself with one question: could a stranger fund this project after 3 sentences?

Resource Kit →



EXTENDED DISCUSSION • 15 MIN

Now the floor is yours

30 extra minutes beyond the core hour: keep writing in Overleaf, or bring me your hardest question.



Submitting

Which venue for my first paper?
What does a rebuttal look like?
How do I answer Reviewer 2?



Collaborating

How do I find co-authors and mentors? How do authorship orders work? Who pays the APC?



Data & ethics

Community consent, anonymisation, and who owns an African dataset once it is on Hugging Face.



Your abstract

Bring me the three sentences you just wrote. I will tell you what a reviewer would say.

No question is too basic. The questions you are afraid to ask are the ones the whole room needs answered.

Resource Kit →





TAKE-HOME

The papers behind this session



The reading method

S. Keshav (2007). "How to Read a Paper." ACM SIGCOMM Computer Communication Review, 37(3):83–84. — the 3-Pass method, 2 pages, read it tonight.



The paper we scanned live

Vaswani et al. (2017). "Attention Is All You Need." NeurIPS 2017. — proceedings.neurips.cc/paper_files/paper/2017



An African AI paper, published

Odounfa, Hounmenou et al. (2025). "Deep learning enables precision agriculture for sustainable chili pepper disease detection in Benin." Discover Artificial Intelligence 5:315 — open access.



An African AI paper, at NeurIPS

Tahi, Houndji, Hounmenou & Glèlè Kakaï (2024). "Using pattern mining to determine fine climatic parameters for maize yield in Benin." IJ-AI 13(4):3930–3941 — first shown at a NeurIPS 2022 workshop.

All of these — plus the checklist, Overleaf templates and dataset-card template — are in your Resource Kit.

Resource Kit →





Master the scientific narrative — so Africa tells its own research stories.

Read 3-Pass method — triage, grasp,
reconstruct

Write Frame · Core · Impact · Honesty

The challenge
Your first submitted paper within 6
months.

Thank you — stay in touch for mentoring & follow-up



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Resource Kit →

